

Dewali Sales Project (1)

May 20, 2026

```
[1]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

1 import csv file

```
[2]: df = pd.read_csv('Diwali Sales Data.csv', encoding= 'unicode_escape')
df
```

```
[2]:
```

	User_ID	Cust_name	Product_ID	Gender	Age	Group	Age	Marital_Status	\
0	1002903	Sanskriti	P00125942	F	26-35	28		0	
1	1000732	Kartik	P00110942	F	26-35	35		1	
2	1001990	Bindu	P00118542	F	26-35	35		1	
3	1001425	Sudevi	P00237842	M	0-17	16		0	
4	1000588	Joni	P00057942	M	26-35	28		1	
...	...	...	...	...	...	...			
11246	1000695	Manning	P00296942	M	18-25	19		1	
11247	1004089	Reichenbach	P00171342	M	26-35	33		0	
11248	1001209	Oshin	P00201342	F	36-45	40		0	
11249	1004023	Noonan	P00059442	M	36-45	37		0	
11250	1002744	Brumley	P00281742	F	18-25	19		0	

	State	Zone	Occupation	Product_Category	Orders	\
0	Maharashtra	Western	Healthcare	Auto	1	
1	Andhra Pradesh	Southern	Govt	Auto	3	
2	Uttar Pradesh	Central	Automobile	Auto	3	
3	Karnataka	Southern	Construction	Auto	2	
4	Gujarat	Western	Food Processing	Auto	2	
...	...	...	...	...	...	
11246	Maharashtra	Western	Chemical	Office	4	
11247	Haryana	Northern	Healthcare	Veterinary	3	
11248	Madhya Pradesh	Central	Textile	Office	4	
11249	Karnataka	Southern	Agriculture	Office	3	
11250	Maharashtra	Western	Healthcare	Office	3	

	Amount	Status	unnamed1
0	23952.0	NaN	NaN
1	23934.0	NaN	NaN
2	23924.0	NaN	NaN
3	23912.0	NaN	NaN
4	23877.0	NaN	NaN
...	...	...	...
11246	370.0	NaN	NaN
11247	367.0	NaN	NaN
11248	213.0	NaN	NaN
11249	206.0	NaN	NaN
11250	188.0	NaN	NaN

[11251 rows x 15 columns]

```
[3]: df.shape
```

```
[3]: (11251, 15)
```

```
[4]: df.head()
```

```
[4]:   User_ID  Cust_name Product_ID Gender Age Group  Age  Marital_Status  \
0  1002903  Sanskriti  P00125942     F   26-35   28             0
1  1000732    Kartik  P00110942     F   26-35   35             1
2  1001990    Bindu  P00118542     F   26-35   35             1
3  1001425    Sudevi  P00237842     M    0-17   16             0
4  1000588     Joni  P00057942     M   26-35   28             1
```

	State	Zone	Occupation	Product_Category	Orders	\
0	Maharashtra	Western	Healthcare	Auto	1	
1	Andhra Pradesh	Southern	Govt	Auto	3	
2	Uttar Pradesh	Central	Automobile	Auto	3	
3	Karnataka	Southern	Construction	Auto	2	
4	Gujarat	Western	Food Processing	Auto	2	

	Amount	Status	unnamed1
0	23952.0	NaN	NaN
1	23934.0	NaN	NaN
2	23924.0	NaN	NaN
3	23912.0	NaN	NaN
4	23877.0	NaN	NaN

```
[5]: df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 11251 entries, 0 to 11250
Data columns (total 15 columns):
#   Column              Non-Null Count  Dtype
#   ...
```

```

---  -----
0    User_ID          11251 non-null  int64
1    Cust_name        11251 non-null  object
2    Product_ID       11251 non-null  object
3    Gender           11251 non-null  object
4    Age Group        11251 non-null  object
5    Age              11251 non-null  int64
6    Marital_Status   11251 non-null  int64
7    State            11251 non-null  object
8    Zone             11251 non-null  object
9    Occupation       11251 non-null  object
10   Product_Category 11251 non-null  object
11   Orders           11251 non-null  int64
12   Amount           11239 non-null  float64
13   Status            0 non-null    float64
14   unnamed1         0 non-null    float64

```

dtypes: float64(3), int64(4), object(8)

memory usage: 1.3+ MB

```

[6]: #drop unrelated /blank columns
import pandas as pd
df.drop(['Status', 'unnamed1'], axis=1, inplace=True)
df

```

```

[6]:      User_ID  Cust_name Product_ID Gender Age Group  Age  Marital_Status  \
0      1002903   Sanskriti  P00125942      F   26-35   28           0
1      1000732     Kartik  P00110942      F   26-35   35           1
2      1001990     Bindu  P00118542      F   26-35   35           1
3      1001425     Sudevi  P00237842      M    0-17   16           0
4      1000588      Joni  P00057942      M   26-35   28           1
...
11246  1000695    Manning  P00296942      M   18-25   19           1
11247  1004089  Reichenbach  P00171342      M   26-35   33           0
11248  1001209     Oshin  P00201342      F   36-45   40           0
11249  1004023    Noonan  P00059442      M   36-45   37           0
11250  1002744    Brumley  P00281742      F   18-25   19           0

      State      Zone      Occupation Product_Category  Orders  \
0  Maharashtra  Western      Healthcare           Auto        1
1  Andhra Pradesh  Southern          Govt           Auto        3
2  Uttar Pradesh  Central      Automobile           Auto        3
3  Karnataka      Southern      Construction           Auto        2
4  Gujarat        Western  Food Processing           Auto        2
...
11246  Maharashtra  Western      Chemical           Office        4
11247  Haryana      Northern      Healthcare      Veterinary        3
11248  Madhya Pradesh  Central      Textile           Office        4

```

11249	Karnataka	Southern	Agriculture	Office	3
11250	Maharashtra	Western	Healthcare	Office	3

	Amount
0	23952.0
1	23934.0
2	23924.0
3	23912.0
4	23877.0
...	...
11246	370.0
11247	367.0
11248	213.0
11249	206.0
11250	188.0

[11251 rows x 13 columns]

```
[7]: # check null values
pd.isnull(df).sum()
```

```
[7]: User_ID          0
Cust_name          0
Product_ID        0
Gender            0
Age Group         0
Age              0
Marital_Status    0
State            0
Zone            0
Occupation       0
Product_Category  0
Orders           0
Amount          12
dtype: int64
```

```
[8]: # delet of null value
df.dropna(inplace=True)
```

```
[9]: # check data type
df['Amount'].dtypes
```

```
[9]: dtype('float64')
```

```
[10]: #change data types
df['Amount'] = df['Amount'].astype('int')
df
```

```
[10]:
```

	User_ID	Cust_name	Product_ID	Gender	Age Group	Age	Marital_Status	\
0	1002903	Sanskriti	P00125942	F	26-35	28		0
1	1000732	Kartik	P00110942	F	26-35	35		1
2	1001990	Bindu	P00118542	F	26-35	35		1
3	1001425	Sudevi	P00237842	M	0-17	16		0
4	1000588	Joni	P00057942	M	26-35	28		1
...	...	...	...	...	...	...		...
11246	1000695	Manning	P00296942	M	18-25	19		1
11247	1004089	Reichenbach	P00171342	M	26-35	33		0
11248	1001209	Oshin	P00201342	F	36-45	40		0
11249	1004023	Noonan	P00059442	M	36-45	37		0
11250	1002744	Brumley	P00281742	F	18-25	19		0

	State	Zone	Occupation	Product_Category	Orders	\
0	Maharashtra	Western	Healthcare	Auto	1	
1	Andhra Pradesh	Southern	Govt	Auto	3	
2	Uttar Pradesh	Central	Automobile	Auto	3	
3	Karnataka	Southern	Construction	Auto	2	
4	Gujarat	Western	Food Processing	Auto	2	
...	...	...	...	...	...	
11246	Maharashtra	Western	Chemical	Office	4	
11247	Haryana	Northern	Healthcare	Veterinary	3	
11248	Madhya Pradesh	Central	Textile	Office	4	
11249	Karnataka	Southern	Agriculture	Office	3	
11250	Maharashtra	Western	Healthcare	Office	3	

	Amount
0	23952
1	23934
2	23924
3	23912
4	23877
...	...
11246	370
11247	367
11248	213
11249	206
11250	188

[11239 rows x 13 columns]

```
[11]: df['Amount'].dtypes
```

```
[11]: dtype('int64')
```

```
[12]: # check total columns
df.columns
```

```
[12]: Index(['User_ID', 'Cust_name', 'Product_ID', 'Gender', 'Age Group', 'Age',
          'Marital_Status', 'State', 'Zone', 'Occupation', 'Product_Category',
          'Orders', 'Amount'],
          dtype='object')
```

```
[13]: # coulmns rename
df.rename (columns = {'Marital_Status':'shaadi'})
```

```
[13]:
```

	User_ID	Cust_name	Product_ID	Gender	Age Group	Age	shaadi	\
0	1002903	Sanskriti	P00125942	F	26-35	28	0	
1	1000732	Kartik	P00110942	F	26-35	35	1	
2	1001990	Bindu	P00118542	F	26-35	35	1	
3	1001425	Sudevi	P00237842	M	0-17	16	0	
4	1000588	Joni	P00057942	M	26-35	28	1	
...	...	...	...	...	...	...	...	...
11246	1000695	Manning	P00296942	M	18-25	19	1	
11247	1004089	Reichenbach	P00171342	M	26-35	33	0	
11248	1001209	Oshin	P00201342	F	36-45	40	0	
11249	1004023	Noonan	P00059442	M	36-45	37	0	
11250	1002744	Brumley	P00281742	F	18-25	19	0	

	State	Zone	Occupation	Product_Category	Orders	\
0	Maharashtra	Western	Healthcare	Auto	1	
1	Andhra Pradesh	Southern	Govt	Auto	3	
2	Uttar Pradesh	Central	Automobile	Auto	3	
3	Karnataka	Southern	Construction	Auto	2	
4	Gujarat	Western	Food Processing	Auto	2	
...	...	...	...	...	...	...
11246	Maharashtra	Western	Chemical	Office	4	
11247	Haryana	Northern	Healthcare	Veterinary	3	
11248	Madhya Pradesh	Central	Textile	Office	4	
11249	Karnataka	Southern	Agriculture	Office	3	
11250	Maharashtra	Western	Healthcare	Office	3	

	Amount
0	23952
1	23934
2	23924
3	23912
4	23877
...	...
11246	370
11247	367
11248	213
11249	206
11250	188

[11239 rows x 13 columns]

```
[14]: # describe() method returns description of the data in the DataFrame (i.e. ↵
      ↪ count, mean, std, etc)
df.describe()
```

```
[14]:
```

	User_ID	Age	Marital_Status	Orders	Amount
count	1.123900e+04	11239.000000	11239.000000	11239.000000	11239.000000
mean	1.003004e+06	35.410357	0.420055	2.489634	9453.610553
std	1.716039e+03	12.753866	0.493589	1.114967	5222.355168
min	1.000001e+06	12.000000	0.000000	1.000000	188.000000
25%	1.001492e+06	27.000000	0.000000	2.000000	5443.000000
50%	1.003064e+06	33.000000	0.000000	2.000000	8109.000000
75%	1.004426e+06	43.000000	1.000000	3.000000	12675.000000
max	1.006040e+06	92.000000	1.000000	4.000000	23952.000000

```
[15]: # use describe() for specific columns
df[['Age', 'Orders', 'Amount']].describe()
```

```
[15]:
```

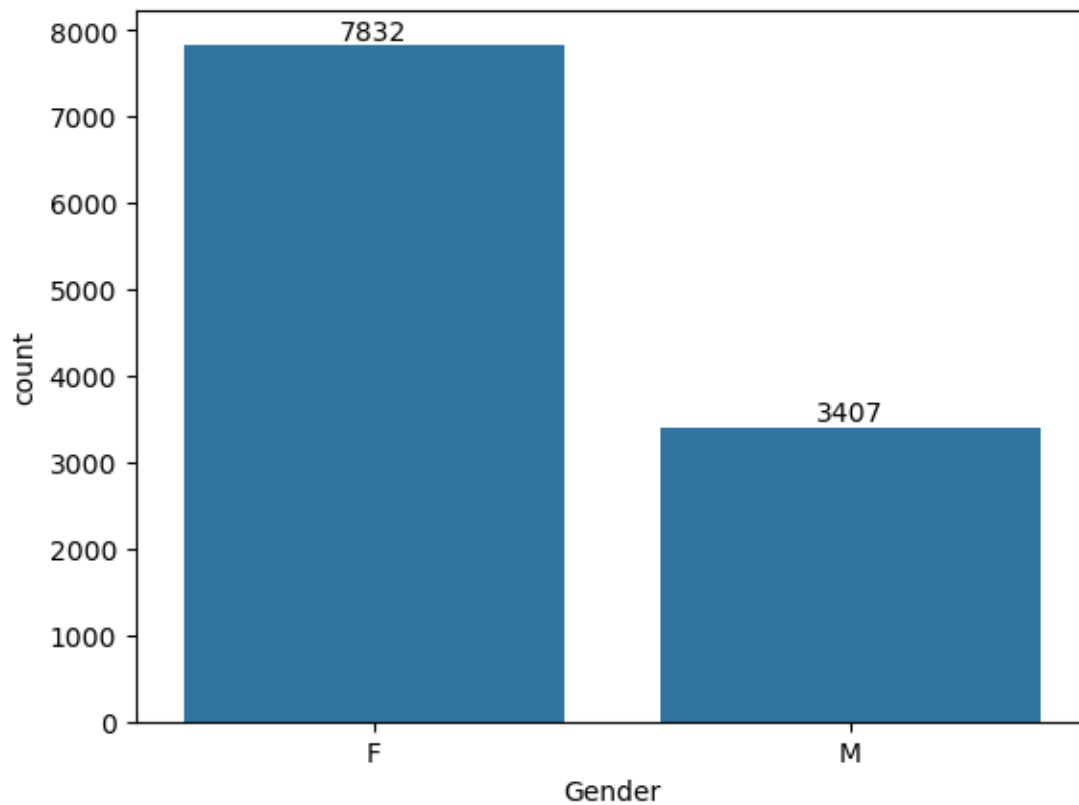
	Age	Orders	Amount
count	11239.000000	11239.000000	11239.000000
mean	35.410357	2.489634	9453.610553
std	12.753866	1.114967	5222.355168
min	12.000000	1.000000	188.000000
25%	27.000000	2.000000	5443.000000
50%	33.000000	2.000000	8109.000000
75%	43.000000	3.000000	12675.000000
max	92.000000	4.000000	23952.000000

Exploratory Data Analysis

Gender

```
[16]: # PLOTTING A BAR GRAPH FOR GENDER AND ITS'S COUNT
ax = sns.countplot(x = 'Gender', data = df)

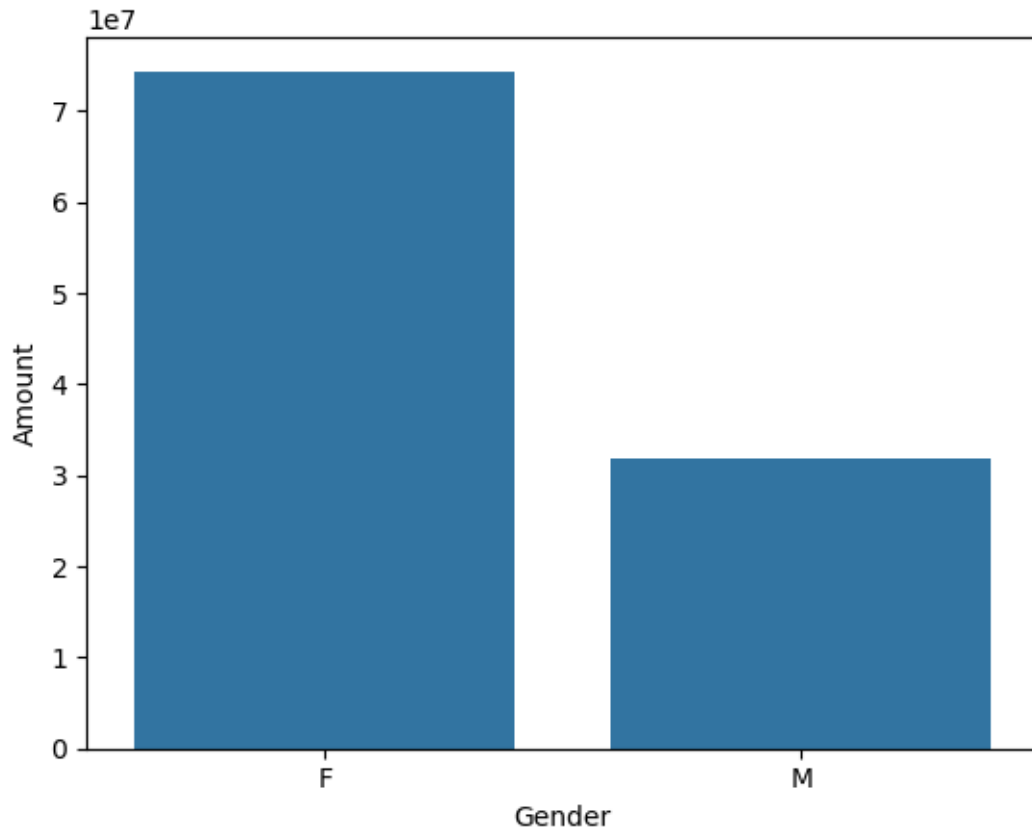
for bars in ax.containers:
    ax.bar_label(bars)
```



```
[17]: # plotting a bar chart for gender vs total amount
sales_gen = df.groupby(['Gender'], as_index=False)['Amount'].sum().
    ↪sort_values(by='Amount', ascending=False)

sns.barplot(x='Gender', y='Amount', data=sales_gen)

plt.show()
```

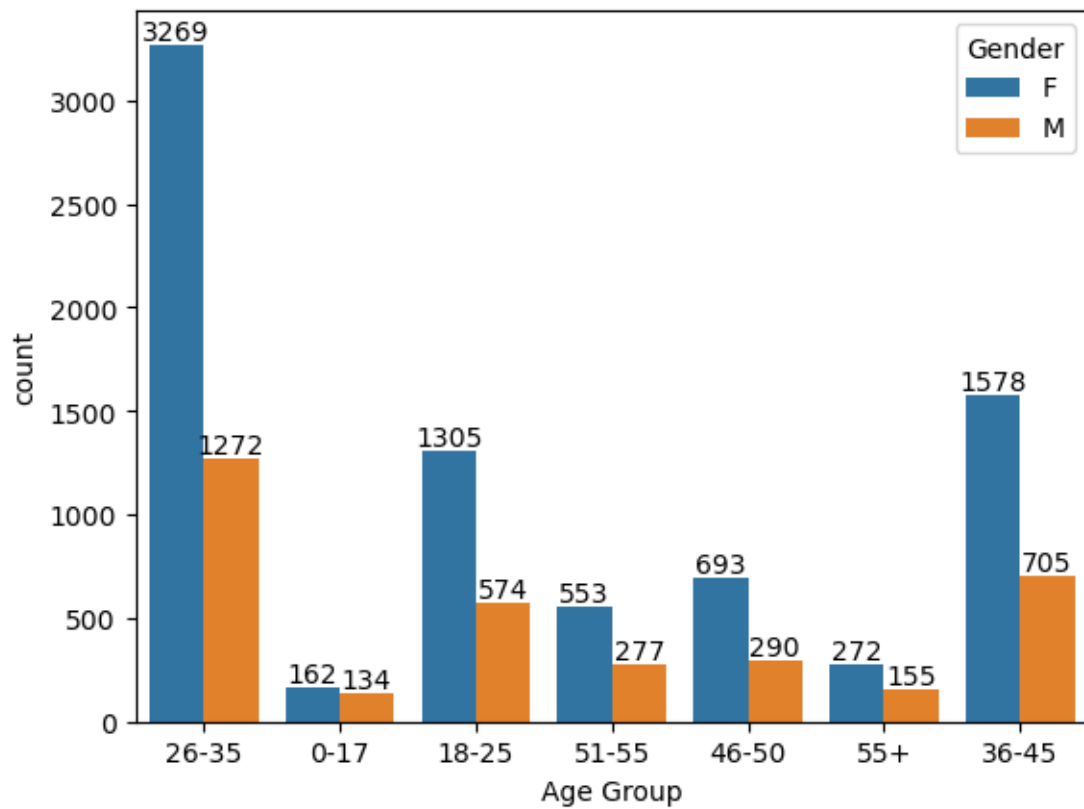



from above graphs we can see that most of the buyers are female and even the purchasing power of females are greater than men

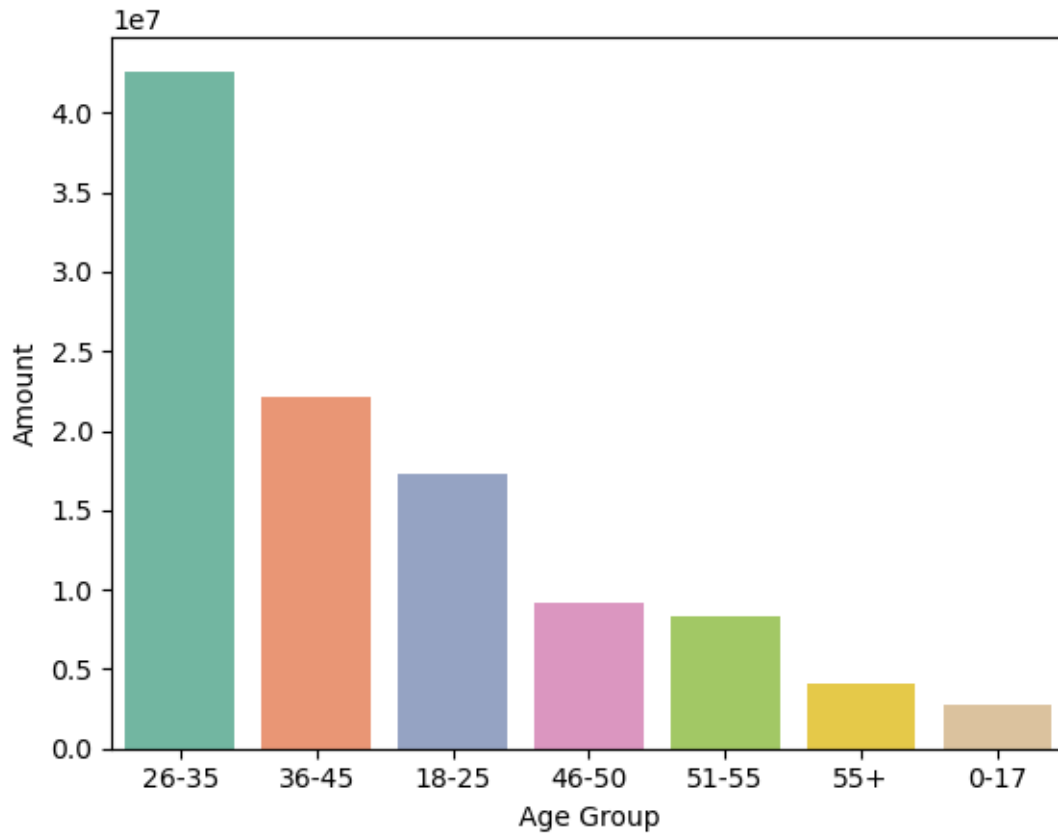
Age

```
[18]: # creating a count plot graph for checking kon category kitni bar repeat hui hai
      ↪ hue= ue graph ko categories ke according color divide karta hai.
ax = sns.countplot(data = df, x = 'Age Group', hue = 'Gender')

for bars in ax.containers:
    ax.bar_label(bars)
```



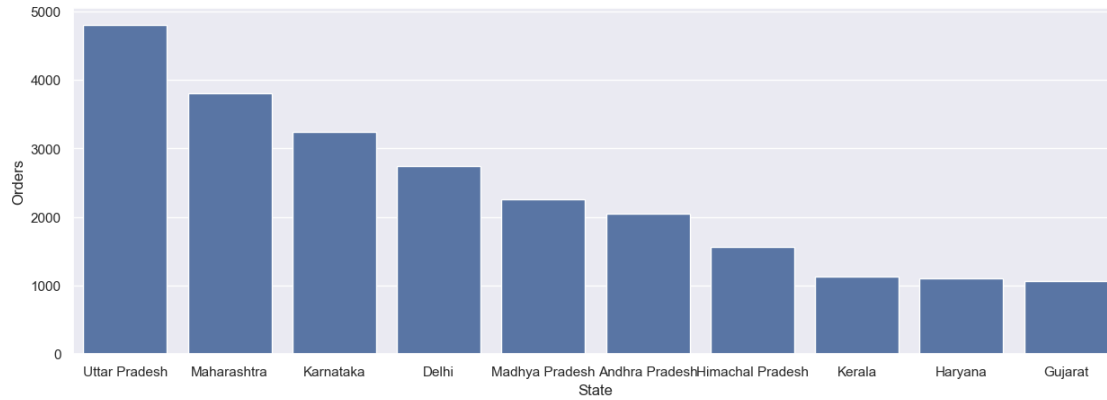
```
[19]: # Total Amount vs Age Group
sales_age = df.groupby(['Age Group'], as_index=False)['Amount'].sum().
    ↪sort_values(by='Amount', ascending=False)
sns.barplot(
    x='Age Group',
    y='Amount',
    data=sales_age,
    hue='Age Group',
    palette='Set2',
    legend=False
)
plt.show()
```



from above graph we can see that most of the buyer are of age group between 26-35 yr female
State

```
[20]: # total number of orders form top 10 state
sales_state= df.groupby (['State'],as_index=False)['Orders'].sum().
↳sort_values(by='Orders',ascending=False). head(10)
sns.set(rc={'figure.figsize':(15,5)})

sns.barplot(data =sales_state,x='State',y='Orders')
plt.show()
```

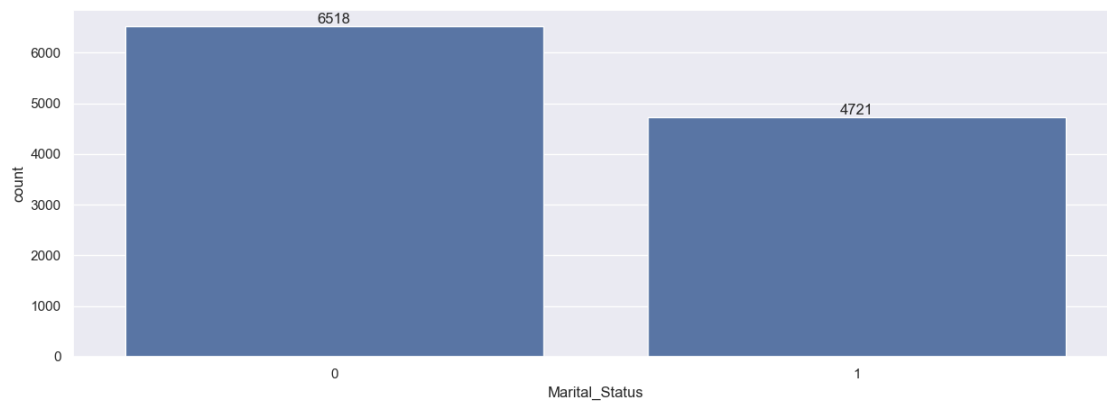


from above graph we can see that most of the order & total sales /amount are from Uttar pradesh, maharashtra, and karnataka respectively

Marital Status

```
[21]: ax = sns.countplot(data = df, x = 'Marital_Status')

sns.set(rc={'figure.figsize':(7,5)})
for bars in ax.containers:
    ax.bar_label(bars)
```



```
[22]: df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
Index: 11239 entries, 0 to 11250
Data columns (total 13 columns):
#   Column      Non-Null Count  Dtype
---  -
0   User_ID     11239 non-null  int64
```

```

1  Cust_name      11239 non-null object
2  Product_ID     11239 non-null object
3  Gender         11239 non-null object
4  Age Group      11239 non-null object
5  Age           11239 non-null int64
6  Marital_Status 11239 non-null int64
7  State         11239 non-null object
8  Zone          11239 non-null object
9  Occupation     11239 non-null object
10 Product_Category 11239 non-null object
11 Orders        11239 non-null int64
12 Amount        11239 non-null int64

```

dtypes: int64(5), object(8)

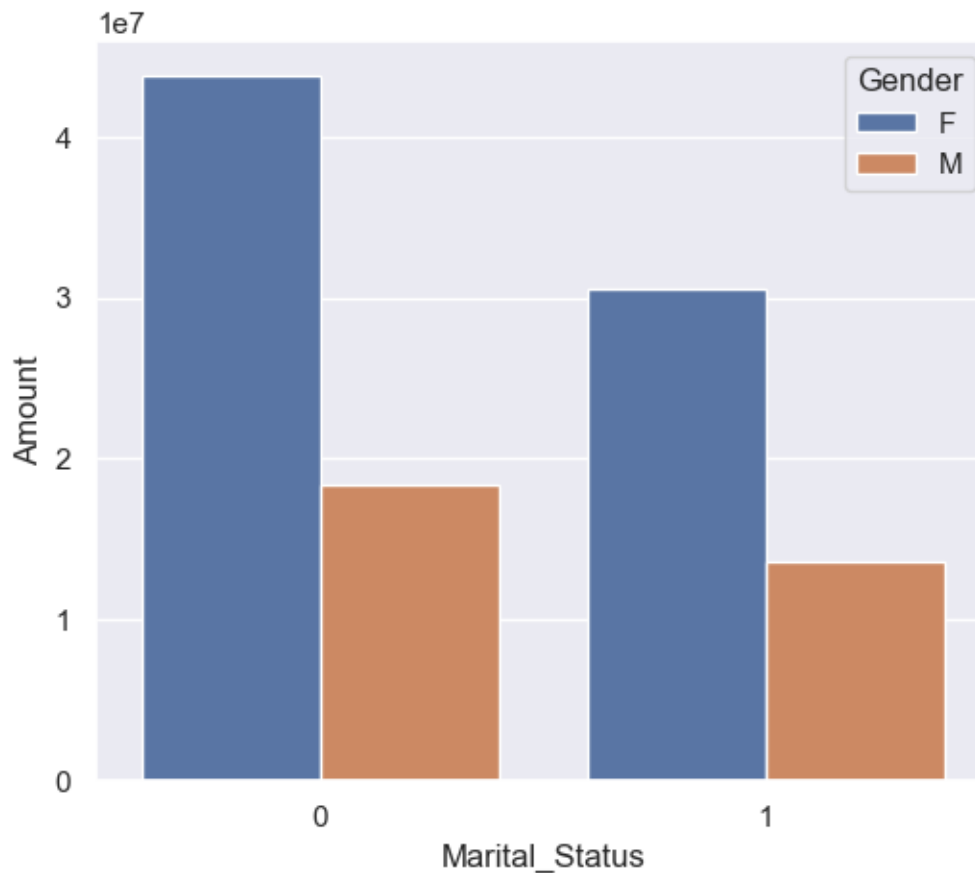
memory usage: 1.2+ MB

```

[23]: sales_state = df.groupby(['Marital_Status', 'Gender'],
    ↪as_index=False)['Amount'].sum().sort_values(by='Amount', ascending=False)

sns.set(rc={'figure.figsize':(6,5)})
sns.barplot(data = sales_state, x = 'Marital_Status',y= 'Amount', hue='Gender')
plt.show()

```

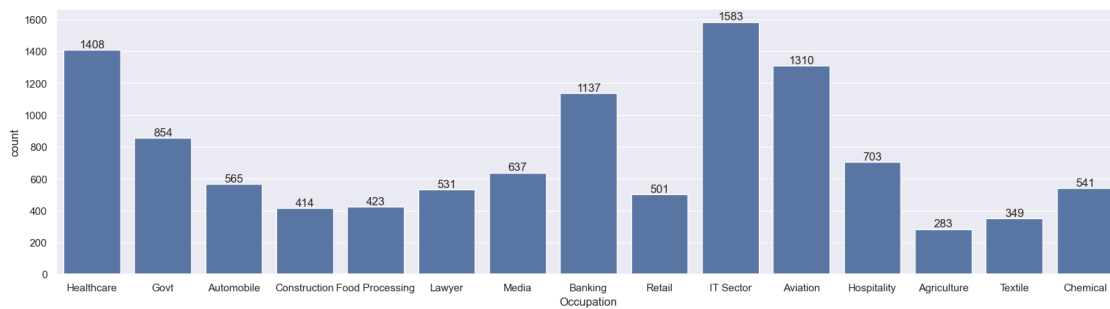


From above graphs we Can see that most of the buyers are married (women) and they have high purchasing Power

Occupation

```
[24]: # Without color
sns.set(rc={'figure.figsize':(20,5)})
ax = sns.countplot(data = df, x = 'Occupation')

for bars in ax.containers:
    ax.bar_label(bars)
```

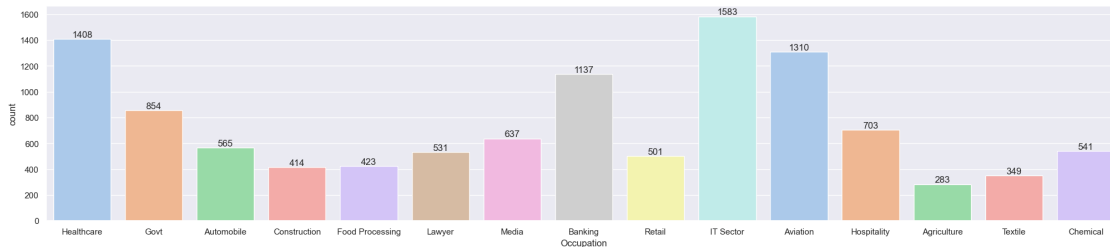


```
[25]: # with color
sns.set(rc={'figure.figsize':(25,5)})

ax = sns.countplot(
    data=df,
    x='Occupation',
    hue='Occupation',
    palette='pastel',
    legend=False
)

for bars in ax.containers:
    ax.bar_label(bars)

plt.show()
```



From above graphs we can see that most of the buyers are working In IT, Healthcare and Aviation sectors

Product category

```
[26]: sns.set(rc={'figure.figsize':(25,5)})

ax = sns.countplot(
    data=df,
    x='Product_Category',
    palette='pastel'
)

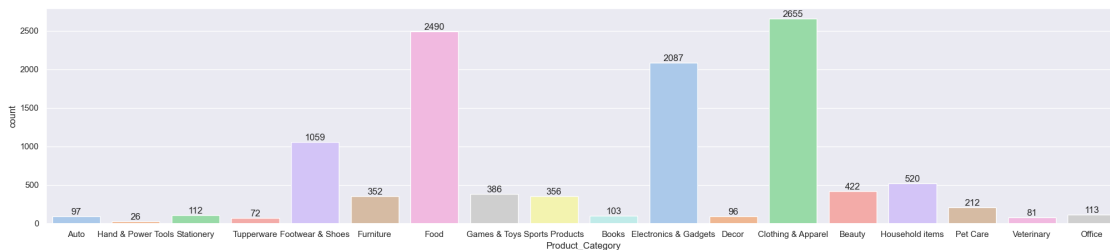
for bars in ax.containers:
    ax.bar_label(bars)

plt.show()
```

C:\Users\pinka\AppData\Local\Temp\ipykernel_6328\4054942876.py:3: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assign the `x` variable to `hue` and set `legend=False` for the same effect.

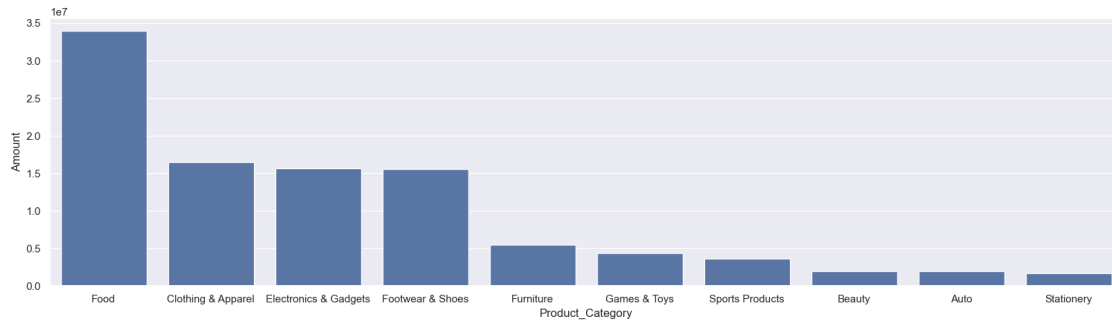
```
ax = sns.countplot(
```



```
[30]: sales_state = df.groupby(['Product_Category'], as_index=False)['Amount'].sum().
        ↪sort_values(by='Amount', ascending=False).head(10)
```

```
sns.set(rc={'figure.figsize':(20,5)})
sns.barplot(data = sales_state, x = 'Product_Category',y= 'Amount')
```

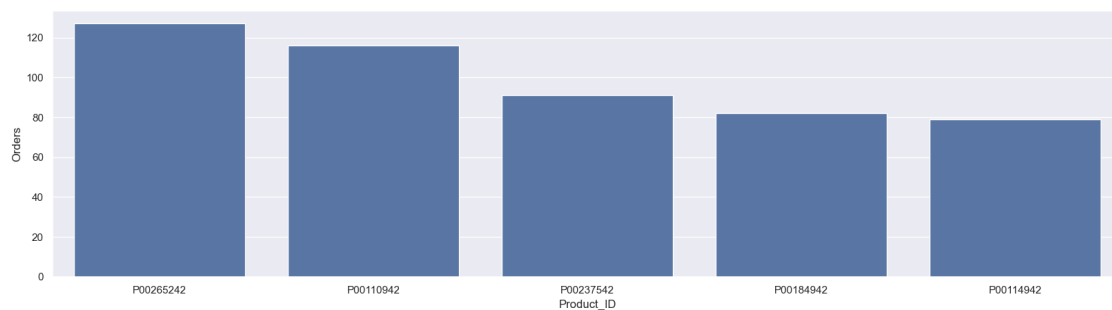
[30]: <Axes: xlabel='Product_Category', ylabel='Amount'>



from above graphs we can see that most of the sold products are from ,Clothing and Electronic Category

```
[28]: sales_state = df.groupby(['Product_ID'], as_index=False)['Orders'].sum().
      ↪sort_values(by='Orders', ascending=False).head()

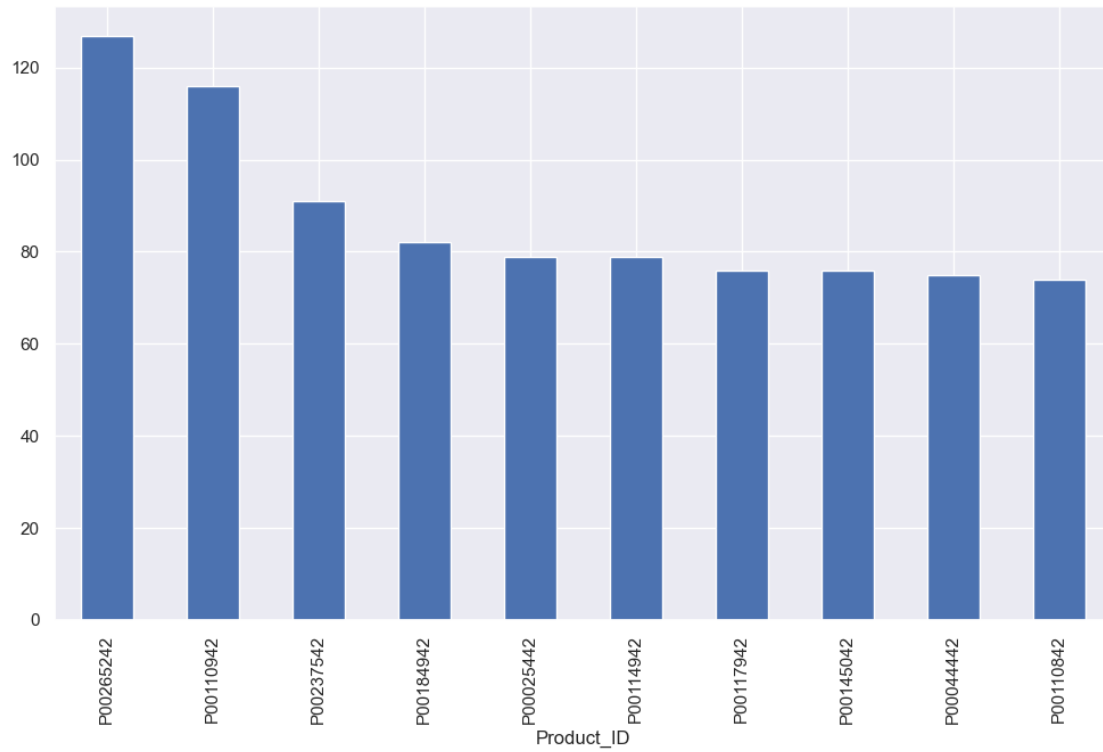
sns.set(rc={'figure.figsize':(20,5)})
sns.barplot(data = sales_state, x = 'Product_ID',y= 'Orders')
plt.show()
```



```
[29]: # top 10 most sold products (same thing as above)

fig1, ax1 = plt.subplots(figsize=(12,7))
df.groupby('Product_ID')['Orders'].sum().nlargest(10).
  ↪sort_values(ascending=False).plot(kind='bar')
```

[29]: <Axes: xlabel='Product_ID'>



Conclusion

Married women age group 26-35 yrs from UP, Maharastra and Karnataka working in IT ,Healthcare and Aviation are more likly to buy product from food,clothing and Electroinc category